HAW HAMBURG

Faculty EMI
Center for Microelectronics and Embedded Software
Microcontroller Lab

# Lab Session 2

# LED pendulum with ultrasonic module

**This Lab Session requires a thorogh preparation!**

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## 1. Objectives of the Laboratory Experiment

In this experiment, you will become familiar with the general-purpose timers of the processor. Your task is to implement the output of patterns and numbers on an LED pendulum, as well as the distance measurement using an ultrasonic sensor.

## 2. Preparation

The Lab session will begin with a written test. You are expected to know the functionalities of the GPIO registers and the steps to configure the ports and pins. You do not need to remember the makro for every register. You will write a few lines of C-Code to configure registers.

To prepare for the lab task, please do the following steps:

- ☐ I have read the experiment instructions thoroughly and have an idea of how to implement them.
- ☐ I have familiarized myself with the lecture content on general-purpose timers.
- ☐ I understand how the LED pendulum and the ultrasonic module works.
- ☐ I have thought about how to configure the GPIO port registers and the general-purpose timer for the tasks.

# 1 Laboratory Tasks

## *1.1 LED Pendulum*

Write a program that shows a narrow vertical bar at the left and right edge of the pendulum display. The switch-on time of the LEDs should be realized with a software delay (for loop). Experiment with the switch-on time of the LEDs, to get an attractive appearance of the bars. Detect the pendulum turnaround time points by continuous polling of the signal L/R.

## *1.2 Ultrasonic Distance Measurement*

Put the ultrasonic module into operation and write a program that performs a continuous distance measurement. Use a general purpose timer for time measurement.

**a.** Oscilloscope the "Trigger" and "Echo" signals and determine the resulting sound travel time for 2 different distances. Sketch the results.

**b.** Use the *printf* command to generate an output of the measured distance on the console.

## *1.3 Display of Measurements on the LED Pendulum*

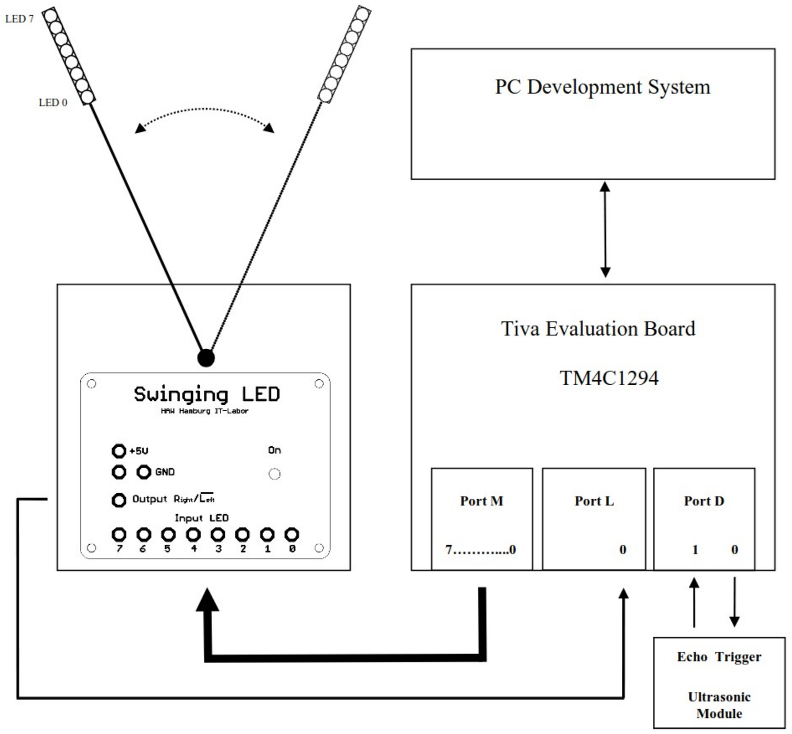
**a.** Generate an output of the determined distance as a horizontal bar on the LED pendulum.


**or**

**b.** Generate an output of the determined distance in the form of a Number output (e.g. "65 cm") on the LED pendulum.


The illustrations in tasks 3.a and 3.b are exemplary. You can create your own display types.
Hint: Use the time of one pendulum direction for measuring and the other for LED output.

# Appendix 1: LED Pendulum Module




- After switching on, the pendulum movement starts with a frequency of approx. 8.3 Hz that cannot be influenced.
- The left reversal point of the pendulum is marked by the rising edge of the signal.
- The right reversal point of the pendulum is marked by the falling edge of the signal.
- The uppermost LED has the name "LED 7" ( High Potential = LED on ).

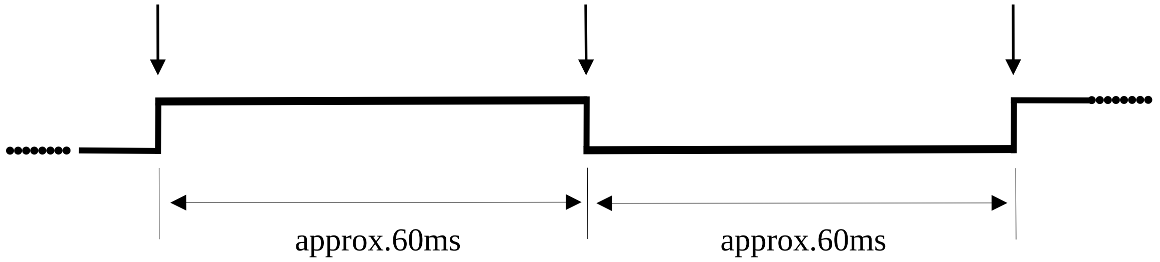



Figure: Signal R/NOT (L)

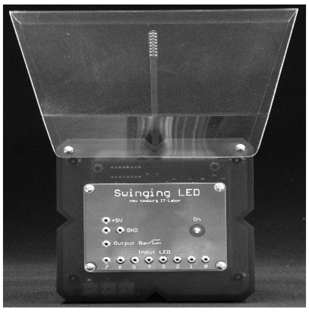


Figure: LED pendulum at rest


Figure: LED pendulum in operation

# Appendix 2: Ultrasonic Module HC-SR04

A distance measurement in the ultrasonic range is triggered via a falling edge of the trigger input "Trigger", started by the user program.

The module generates a positive edge at the echo output "Echo", when the measurement starts. A falling edge at the echo output "Echo" marks the end of the measurement. The time between these two edges corresponds to twice the transit time of the ultrasonic pulse. With the help of knowledge of the ultrasound propagation speed, the distance can be calculated.

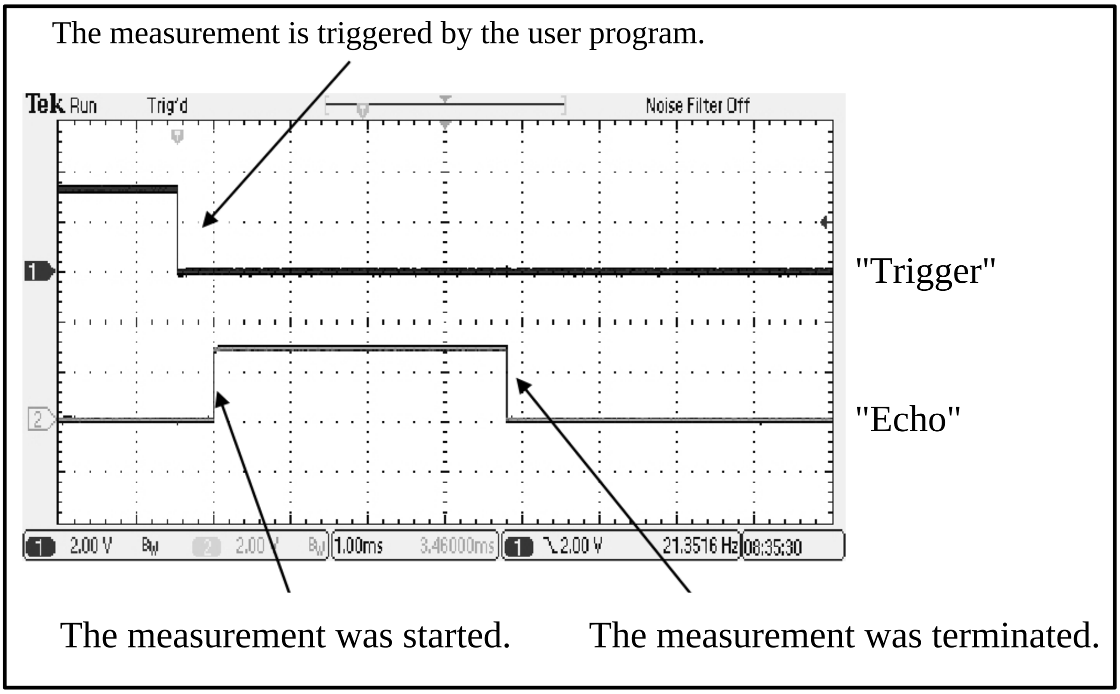




Figure: Timing of the ultrasonic module signals at a distance of ~65 cm.